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Datasci 224 Final Project

Support Vector Machines for Predicting Incident Parkinson's Disease Using Deep Learning Sleep Features

Background

Parkinson's disease (PD) is a neurodegenerative disease that affects one's movement and balance. Around 1.5 million Americans are affected, and there are 90,000 new diagnoses every year. Incidence rises with age, with symptoms typically emerging around age 60, and the condition occurs more frequently in men.² PD also involves a range of other features, including sleep disturbances and cognitive decline, many of which appear earlier than motor symptoms. It is crucial to identify individuals at risk early, as clinical diagnosis typically occurs only after substantial neurodegeneration.

Ganglberger et al. have been developing a deep-learning-based Brain Health Score (BHS) derived from polysomnographic (PSG) sleep EEG recordings. The BHS captures many patterns in sleep architecture that reflect underlying brain health, providing a non-invasive, highly interpretable biomarker.¹ While our primary analysis examined the association between the scalar BHS and incident PD using traditional survival methods, the deep learning model that generates the BHS also produces a 1,024-dimensional internal representation at the last hidden layer (LHL) before compressing it into the final score. For a secondary analysis, we applied a Support Vector Machine (SVM) classifier to evaluate whether the 1,024-dimensional LHL feature vector improves PD prediction compared to the scalar brain health score.

Dataset

This is a prospective cohort analysis using data from the Osteoporotic Fractures in Men Study (MrOS), which consists of community-dwelling men aged 65 years and older enrolled at six US clinical sites. The Brain Health Score and the 1,024-dimensional LHL feature vector were developed by Ganglberger et al. using a deep learning model trained on PSG recordings from six cohorts. For this analysis, BHS values were generated by applying this pre-trained model to PSG recordings obtained at the MrOS sleep visit and linked to longitudinal follow-up data on PD incidence.¹ The primary outcome was incident PD, which is new PD diagnosis occurring after the baseline sleep visit. Seven baseline covariates were included based on the primary analysis: age at sleep visit, race, BMI, physical activity score, diabetes, hypertension, and education. Participants with prevalent PD at baseline were excluded to look at incident-only PD. Participants with missing outcome data were also excluded, yielding a final sample of 2,714

men. 62 of these men developed incident PD during follow-up, which is approximately 2.3% of the population.

Table 1. Baseline characteristics of MrOS participants by Brain Health Score tertile

Values are mean (SD) for continuous variables and n (%) for categorical variables.

	Low BHS (n=905)	Medium BHS (n=904)	High BHS (n=905)	p-value
Demographics				
Age at sleep visit (years)	78.11 (5.71)	76.30 (5.25)	74.11 (4.38)	<0.001
Non-white race, n (%)	68 (7.5%)	82 (9.1%)	81 (9.0%)	0.418
Education (1–8 scale)	5.79 (1.62)	5.99 (1.66)	6.11 (1.53)	<0.001
Clinical characteristics				
BMI (kg/m ²)	27.80 (4.04)	27.19 (3.74)	26.62 (3.39)	<0.001
Physical activity (PASE score)	138.85 (71.88)	150.61 (72.71)	153.94 (68.25)	<0.001
Diabetes, n (%)	145 (16.0%)	108 (11.9%)	99 (10.9%)	0.003
Hypertension, n (%)	490 (54.1%)	459 (50.8%)	395 (43.6%)	<0.001
Descriptive variables†				
Depression (GDS ≥ 6), n (%)	71 (7.8%)	46 (5.1%)	42 (4.6%)	0.007
Cognitive function (3MS score)	92.08 (5.89)	93.36 (4.99)	94.15 (4.76)	<0.001
Outcome				
Incident PD, n (%)	32 (3.5%)	16 (1.8%)	14 (1.5%)	0.008

BHS = Brain Health Score; GDS = Geriatric Depression Scale; 3MS = Modified Mini-Mental State Examination; PASE = Physical Activity Scale for the Elderly; PD = Parkinson's disease.

P-values from ANOVA for continuous variables and chi-squared tests for categorical variables across BHS tertiles.

† Depression and cognitive function are included for descriptive purposes and were not included in the primary model due to concerns about overadjustment; see sensitivity analyses.

SVM: Machine Learning Algorithm

Support Vector Machines (SVMs) are supervised classification algorithms. There are several reasons why I chose SVM for this project. First, the feature space is high-dimensional with 1,031 features (7 covariates + 1,024 LHL dimensions). SVMs are known to perform well in high-dimensional spaces because the data naturally spans a large space, making linear separation more feasible. Second, the dataset has a small case count relative to the number of features, so we have a small n, large p problem. With only 62 PD cases and 1,031 features, overfitting is a serious concern. SVM addresses this naturally because maximizing the margin acts as a built-in regularizer, finding the boundary with the most breathing room between the two classes, which tends to generalize better to unseen data than boundaries that tightly fit the training data. Finally, SVM is designed for binary classification, which is perfect for this situation and research problem.

The hard margin assumes that the two groups are linearly separable. However, data can be messy in reality and hard to separate perfectly. Thus, we use the soft margin to help make the assumption more flexible by allowing for some error and misclassification.

How does SVM work?

The goal of SVM is to create a hyperplane that separates two classes by the maximum possible margin. This margin is defined by the nearest points, also known as the support vectors, for each class/group. These support vectors are the only training points that determine the position of the hyperplane/boundary. The hyperplane is defined as $w^T x_i + b = 0$, where w is the weight vector, x is the patients' feature vectors, and b is the bias. For my data, a patient is classified as PD if $w^T x - b > 0$ and no PD if $w^T x - b < 0$.

The two margin boundaries are defined as $w^T x_i + b = 1$ and $w^T x_i + b = -1$. The margin width is the distance between two parallel planes: $|c_1 - c_2| / \|w\| = |1 - (-1)| / \|w\| = 2 / \|w\|$ where $\|w\|$ is the length of w . To maximize the margin, we minimize the denominator, $\|w\|$ in this case. In order to make it easier to derive, we set it to $1/2 \|w\|^2$ so we can get a clean w . We use the constraint $y_i(w^T x_i + b) \geq 1$ for all i to make sure that all patients are correctly classified outside the margin. Multiplying by y_i ensures that correct classifications always yield a positive value satisfying the constraint.

However, real data is messy and is not always easy to separate into two groups. Thus, there is a soft margin to account for flexibility and violations. It introduces a slack variable $\zeta_i \geq 0$ for each patient that measures the violation with the hinge loss $\zeta_i = \max(0, 1 - y_i(w^T x_i - b))$. When $\zeta_i = 0$, the patient is correctly classified outside the margin, and $\zeta_i > 0$ is proportional to how far inside the margin, or across the boundary, the patient falls. It also introduces a penalty C , where the goal is to minimize $\|w\|^2 + C \cdot (1/n) \sum \zeta_i$ that is constrained to $y_i(w^T x_i - b) \geq 1 - \zeta_i$, $\zeta_i \geq 0$ for all i . A larger C has stricter penalties for violations, while a smaller C allows more violations.

We used a linear kernel. This is appropriate given the already high-dimensional feature space (1,031 features), where a linear boundary is typically sufficient for good separation.

Results

Two linear SVM models were fit and compared using 10-fold stratified cross-validation. For model 1, our baseline, we used 7 covariates and the brain health score with a total of 8 features. For model 2, our model that includes the last hidden layer, there are 7 covariates and 1,024 LHL features, a total of 1,031 features. Features were standardized within each training fold using the fold's mean and standard deviation, which were then applied to the test fold to prevent data leakage. We evaluated the models' performance using the area under the ROC curve (AUC).

Table 2. Linear SVM model performance for predicting incident Parkinson's disease

AUC-ROC estimated from 10-fold stratified cross-validation.

	AUC-ROC	Δ AUC-ROC
Covariates + BHS scalar	0.591	<i>reference</i>
Covariates + 1,024 LHL features	0.611	+0.020
<i>Chance level</i>	<i>0.500</i>	—

AUC-ROC = area under the receiver operating characteristic curve; BHS = Brain Health Score; LHL = last hidden layer; SVM = support vector machine.

Features standardized within each training fold. LinearSVC with $C = 1$ and balanced class weights.

Discussion

The LHL model, with an AUC of 0.611, outperformed the BHS model, with an AUC of 0.591. This suggests that information is lost when compressing to a single scalar score. However, both models performed above the chance level of 0.5.

There are several limitations we have to consider. First, because there are only 62 PD cases, the statistical power is limited. Second, we used the default $C = 1$ and did no hyperparameter tuning beyond that. Further hyperparameter tuning could improve performance. Third, the difference between the two AUCs is very modest and should be interpreted with caution. In future research, a bigger sample size can confirm whether the LHL embedding provides a clinically meaningful improvement.

References

1. Ganglberger W, Sun H, Turley N, Tripathi A, Hadar P, Gupta A, Gallagher K, et al. Brain Health from Sleep EEG: A Multicohort, Deep Learning Biomarker for Cognition, Disease, and Mortality. NEJM AI. 2026;3(3). DOI: 10.1056/AIoa2500487.
2. Cleveland Clinic. Parkinson's Disease: An Overview. Cleveland Clinic; 2024. Available from: <https://my.clevelandclinic.org/health/diseases/8525-parkinsons-disease-an-overview>. Accessed May 2026.